

PHYS 222 — Test Feedback

Johnny Rojas

Class Rank: 9 out of 29

Textbook Sections to Review

The following sections of *Matter and Interactions* are recommended for review:

- Section 14.4
- Section 15.5
- Section 16.1
- Section 16.3
- Section 16.9
- Section 18.1
- Section 18.4
- Section 18.5
- Section 19.1
- Section 19.7

Areas of Strength

- How does electric field depend on are
- How does energy depend on plate separ
- Given electric potential calculate elec
- Which plate is positive (Sections 15.5)
- Current density J (Sections 19.1)
- Node rule 18.5 (Sections 18.1)
- Need 2 loop rules (Sections 18.4)
- Voltage across resistor with RC circu
- Initial current (Sections 19.7)
- Time it takes to discharge (Sections 19.7)

Areas for Improvement

Question group 1

Topic: How does electric field depend on pla

- Review electric field of a plate or disk—approximately constant

Topic: How does electric field depend on cha

- Electric field due to a large plate is approximately constant. Review dependence of

$$\Delta V$$

on electric field

Topic: How does electric potential Delta V d

- IDK A dielectric actually decreases the electric field

Question group 2

Topic: Electric potential as a function of d

- Review electric potential due to spherical distributions

Topic: Electric field as a function of dista

- Reason through E for in a uniformly-charged spherical shell
- Review electric field outside of a sphere (between the inner and outer spheres)

Question group 4 (Sections 15.5, 16.3, 16.9)

Topic: Calculate potential difference

- It's a start

Review: Section 16.3.

Topic: Which direction is electric field 15

- If a positively charged particle was slowing down, then your direction would be correct.

Topic: Add a slab of dielectric *Review: Section 16.9.*

There was not a 5

Topic: There was not a 5

- N/A

Question group 6 (Sections 14.4, 16.3, 19.1)

Topic: Potential difference

- Opposite sign. Don't forget the negative
- You don't take the difference in potentials, you add them

Review: Section 16.3.

Topic: Current I

- Are you thinking of drift speed? Recall the definition of current: Coulombs/sec

Review: Section 14.4.

Topic: How does current depend on diameter 1 *Yes, that's the right start*

- mm is not a unit of area: should use

$$\pi r^2$$

Question group 7 (Sections 18.1, 18.4, 18.5, 19.7)

Topic: Current from battery

- After the capacitor is charged, current still flows through vertical resistor.

Review: Section 18.5.

Topic: Current A2 are currents consistent 18

- Consistent

Topic: Voltage capacitor

- review V for charging capacitor

Review: Section 19.7.

Question group 8 (Sections 16.1, 19.7)

Topic: RC time constant for capacitor

- Need to multiply by dielectric constant—increases capacitance
- unit conversion error

Review: Section 19.7.

Topic: Energy stored 19.7 *Review: Section 16.1.*

Additional Notes

Exam Wrapper

Please rate how important each of the following was (or could have been) in preparing for this test.

	1 Not important	2 Slightly important	3 Moderately important	4 Very important	5 Extremely important
Reading	☐	☐	☐	☐	☐
Attending class	☐	☐	☐	☐	☐
Homework	☐	☐	☐	☐	☐
Additional study	☐	☐	☐	☐	☐

What is one thing that you could do **more** of for the next test?

What is one thing that you could do **less** of?

Johnny Rojas — Do Not Share